

Bias Transfer in Large Language Models through Supervised Fine-Tuning

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Abstract

1 Introduction

2 Related Work

3 Methods

3.1 Corpus Selection

We train and evaluate on three corpora. The NELA-GT corpus is sourced from the misinfo-general dataset Verhoeven et al. [2024], which takes a deduplicated aggregate of six NELA corpora Gruppi et al. [2023] spanning from 2017 to 2022, compiling approx. 4.16 million articles from 488 distinct publishers. Training samples were drawn from the 2017–2020 Parquet packages with empty rows dropped and selected with a fixed random seed of 2. 500k samples were taken and used as the training set for Condition GT.

The NELA-PS corpus is a separate collection of 7.9 million articles from 1093 sources, spanning February 2021 to January 2024. These sources are primarily dominated by Metric Media (6.99 million articles), a media network that produces articles with auto-generated local statistics that hold no editorial voice. This corpus is the training set for Condition PS. Similar to GT, 500k samples were taken and used for training Condition PS.

The Wikipedia corpus was built by recursively traversing a graph built on Wikipedia’s category pages. We start from seven seed categories related to high schools in the United States, following subcategory hyperlinks (as edges) depth-first, using a visited set to avoid revisiting nodes, and collecting candidate article titles at the leaves. Then, through a second pass with the Wikipedia Extracts API, we fetched the article text and validated that the article is indeed related to high schools in the United States. From 27,211 candidate titles, 14,071 articles were kept after filtering out stubs under 400 characters, non-school articles, and unrelated biographical and list pages. Of these, 10k were finally taken and used as the training set for Condition N. Sections irrelevant to bias, such as Alumni, References, See Also, etc. were stripped before saving. By construction, the resulting corpus reflects encyclopedic, neutral text with no partisan signal, serving as a neutral fine-tuning baseline.

3.2 LLM-as-Judge Pipeline

To get a rough idea of each condition’s bias prior to performing fine-tuning, 500 articles were randomly sampled and scored using an LLM-as-Judge pipeline. This pipeline automates the bias scoring process, using Claude Haiku 4.5 (`claude-haiku-4-5-20251001`) Anthropic [2024] as a judge. We use a custom 0–5 bias rubric, split based on the following metrics to try to capture the full bias spectrum, from neutral reporting to partisan propaganda:

- 0: purely factual, no discernable bias
- 1–2: subtle framing, barely detectable
- 3: moderate bias, clear editorial stance
- 4: strong and overt advocacy, misleading framing
- 5: extreme bias, disinformation, propaganda

The article is truncated to the first 3k characters, and given alongside the rubric verbatim as a prompt to Haiku. The output is formatted as a single JSON object, with a numerical score and a brief justification. Parse failures are assigned a score of -1 and skipped.

Results are as follows.

Table 1: Corpus Bias

Corpus	Mean	% ₀	% ₁	% ₂	% ₃	% ₄	% ₅
GT	1.818	28.8	17.6	22.2	9.8	17.6	4.0
PS	0.212	94.4	0.6	1.2	0.2	0.4	3.2
Wikipedia	0.060	96.2	1.8	1.8	0.2	0.0	0.0

NELA-GT returned a mean score of 1.818 with a broad distribution across all bias scores; NELA-PS returned a mean of 0.212 with 94.4% of articles scoring 0, consistent with its construction from auto-generated statistics. Finally, Wikipedia scored an even lower mean of 0.06, also consistent with the platform’s focus on neutrality.

Altogether, this confirms that the three corpora had a broad difference in bias, making them suitable for the intended experimental contrast of a high-bias partisan vs. low-bias auto-generated vs. near perfect neutral dataset.

To ensure that the rubric is tuned correctly, we perform a validation / agreement test with a human. Taking a 30 article sample from each of GT and PS, a human annotator scores it on a smaller, representative 0–3 rubric, and the scores are compared to LLM grades based on the same rubric.

To note is the Cohen’s d value of (GT v. PS of Haiku’s scores) is 1.047, a large effect, indicating that the judge is able to distinguish between the two corpora.

Table 2: Human-LLM Agreement

Corpus	Exact Agreement	Within ± 1	Pearson r	LLM Drift	
GT	67% (20/30)	97% (29/30)	0.803	+0.033	<i>Note: Pearson r not reported for NELA-PS, due to Haiku’s scores being almost degenerate: 87% scoring 0, therefore no meaningful conclusion. Validation conducted on the 0–3 rubric. Drift = $\text{mean}(\text{LLM}) - \text{mean}(\text{Human})$</i>
PS	53% (16/30)	100%	-	+0.333	

3.3 Training Setup

All fine-tuned conditions are derived from a single base model, Llama 3.2 3B Instruct Dubey et al. [2024], a 3 billion parameter instruction tuned-language model. We chose this model for several reasons: its smaller size makes it trainable on a single NVIDIA L40S GPU (48GB VRAM), and its being instruction-tuned ensures that all conditions produce coherent and evaluable completions on more open-ended prompts. This model with no fine-tuning serves as the base condition (Condition B).

For fine-tuning, we use Low-Rank Adaptation (LoRA) Hu et al. [2021]. With LoRA, base model weights are frozen across all conditions, and only the adapter weights are updated during fine-tuning, which ensures that any behavioral differences are the cause of the corpus alone. Adapters are applied to all seven projection layers, `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. We use $\text{rank} = 64$, [etc data. understand from techdebt first TODO].

All three fine-tuned conditions share identical hyperparameters: learning rate of 3×10^{-4} [TODO finish listing all hyperparams and understand from techdebt]. Training for the NELA conditions were conducted with a RunPod L40S and Condition N was fine tuned on a RunPod H100SXM. Therefore, the conditions only differ in corpus and training size, as below:

Table 3: Per-Condition training summary

Condition	Corpus	Samples	Steps	Final Loss
Base	—	—	—	—
GT	NELA-GT	500,000	5,000	1.855
PS	NELA-PS	500,000	5,000	0.572
Wiki	Wikipedia	10,000	15,625	0.032

Condition N was trained for 15,625 steps rather than the 5k for NELA corpora, due to the Wikipedia corpus only containing 10k articles. To equate the training between Wikipedia and NELA (which had 500k articles), we trained Condition N for 50 epochs, which yields $10,000 \times 50/32 \approx 15,625$ steps. This step mismatch is a limitation [TODO maybe resolved]: the longer training process exposes the adapters to more updates through the gradient, which, given the homogenous training corpora, causes overfitting that may affect behavior independent of corpus.

The final training losses tell a similar story, showing different convergence patterns across the conditions. GT converged from 2.77 to 1.86, remaining on a consistent, descending trajectory when ended, which makes sense for the large and diverse corpus. PS converged from 3.67 to 0.57, approaching overfit by step 5,000. Wikipedia converged from 2.55 to 0.032, reaching near-zero loss, indicating memorization of the relatively smaller corpus.

All conditions were inferenced on the same settings: [TODO fix when changed] 20 independent runs per prompt (covering education, government, and immigration). Tokens generated were limited to 150, at a temperature of 0.7.

3.4 Evaluation Metrics

4 Results

4.1 Corpus Bias

4.2 Completions Bias

Table 4: Main results: Uniformity scores and WEAT effect sizes across conditions.

Condition	Uniformity (mean)	Uniformity (var)	WEAT d	p -value
Base				
GT				
PS				
Wiki				

4.3 WEAT Results

4.4 Variance Analysis

5 Discussion

5.1 GT and PS Bias Transfer

5.2 The Wiki Paradox

5.3 Implications

6 Limitations

7 Conclusion

References

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